

Develop a predictive text system using N-Gram and HMM (Hidden Markov Models) to predict the next word based on previous context.

Dataset: Brown Corpus or Wikipedia dataset.

N-gram:

Code:

```
import nltk
from nltk.corpus import brown
from nltk.util import ngrams
from collections import defaultdict, Counter

nltk.download('brown')
words = [word.lower() for word in brown.words()]
bigrams = list(ngrams(words, 2))
model = defaultdict(Counter)

for w1, w2 in bigrams:
    model[w1][w2] += 1

def predict_next(word):
    if word in model:
        return model[word].most_common(5)
    else:
        return "Word not in corpus"

print(predict_next("the"))
```

Output:

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Package brown is already up-to-date!
[('first', 662), ('same', 628), ('most', 417), ('other', 416), ('', 405)]
```

HMM

Code:

```
import nltk
from nltk.corpus import brown
from nltk.tag import hmm

nltk.download('brown')
nltk.download('universal_tagset')

tagged_sents = brown.tagged_sents(tagset='universal')
train_data = tagged_sents[:4000]
test_data = tagged_sents[4000:4500]

trainer = hmm.HiddenMarkovModelTrainer()

hmm_tagger = trainer.train(train_data)

print(hmm_tagger.evaluate(test_data))
```

```
def predict_next_hmm(sentence):
    tokens = sentence.lower().split()
    tags = hmm_tagger.tag(tokens)
    last_tag = tags[-1][1]
    candidates = [word for word, tag in brown.tagged_words(tagset='universal') if tag == last_tag]

    return candidates[:5]

print(predict_next_hmm("the cat"))
```

Output:

```
[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data] Package brown is already up-to-date!
[nltk_data] Downloading package universal_tagset to /root/nltk_data...
[nltk_data] Unzipping taggers/universal_tagset.zip.
/tmp/ipykernel_289/866183440.py:16: DeprecationWarning:
  Function evaluate() has been deprecated. Use accuracy(gold)
  instead.
  print(hmm_tagger.evaluate(test_data))
/usr/local/lib/python3.12/dist-packages/nltk/tag/hmm.py:335: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(si, self._symbols[k])
/usr/local/lib/python3.12/dist-packages/nltk/tag/hmm.py:333: RuntimeWarning: overflow encountered in cast
  X[i, j] = self._transitions[si].logprob(self._states[j])
/usr/local/lib/python3.12/dist-packages/nltk/tag/hmm.py:363: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(si, self._symbols[k])
0.3632816217759589
```

```
['The', 'an', 'no', 'any', 'The']
```